

Graph Diffusion History Reconstruction via Feasibility-Aware Markov Chain Monte Carlo Estimation

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HIGHLIGHTS

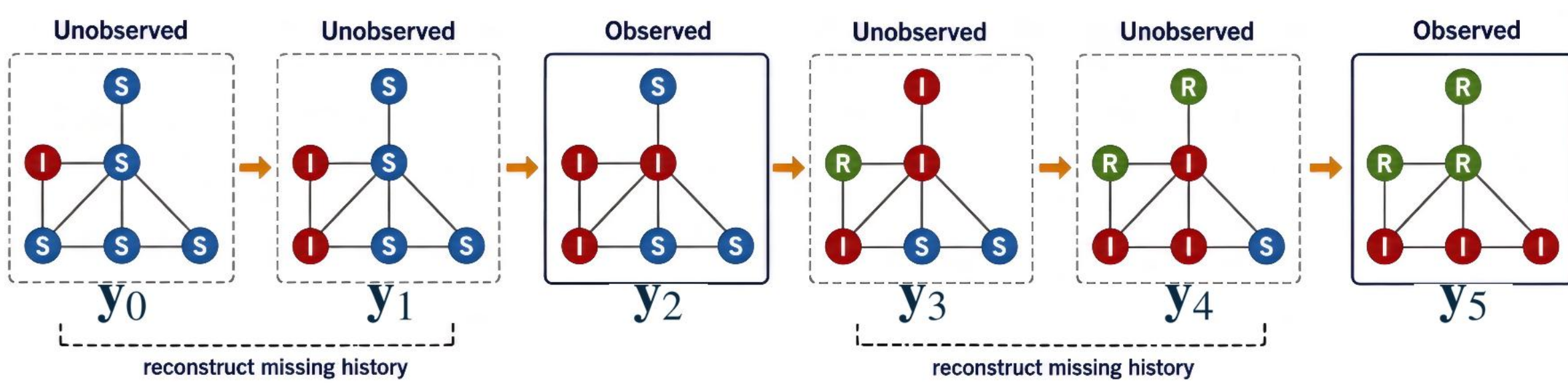
- **Problem:** *multi-snapshot graph diffusion history reconstruction.* Reconstruct the **complete hidden trajectory** from **sparse observed snapshots**. No true parameters or complete training histories are required.
- **Challenge:** sparse observations induce **hard combinatorial constraints**. Exponential history space + segment-wise feasibility across observed times.
- **HERMES:** two-stage reconstruction pipeline. Estimate diffusion parameters by **segmented mean-field pseudolikelihood**. Reduce reconstruction to posterior expected hitting-time estimation.
- **Core method:** *feasibility-aware M–H MCMC.* Dynamic reachability keeps proposals **compatible with all observed snapshots**. A GNN proposal helps MCMC explore likely histories more efficiently.

INTRODUCTION

Diffusion on Graphs



Problem Definition



$\mathcal{T}_{\text{obs}} = \{t_1 < \dots < t_K = T\}$, $\mathbf{y}_{\mathcal{T}_{\text{obs}}} = \{\mathbf{y}_{t_k}\}_{k=1}^K$, $\mathbf{Y} = (\mathbf{y}_0, \dots, \mathbf{y}_T)^\top$

S Susceptible I Infected R Recovered

Challenges

Explosive Search Space
Exponentially many hidden-state assignments.

Coupled Feasibility
Each hidden segment must connect observed endpoints.

Scarcity of Training Data
Complete histories are rarely available as labels.

Previous Methods & Limitations

Supervised Imputation
Requires complete training histories; may ignore diffusion feasibility.

Likelihood-Based Methods
Exact snapshot likelihood is intractable.

Single-Snapshot Methods
Uses only the final snapshot $\mathbf{y}_T$ and discards intermediate observations.

DIFFUSION PARAMETER ESTIMATION

Segmented mean-field pseudolikelihood

Exact snapshot likelihood marginalizes over exponentially many hidden histories and is intractable. We split the timeline at observed snapshots and propagate node-wise mean-field marginals independently within each segment.



The first segment starts from the initial prior; each later segment is initialized by its observed left endpoint.

Mean-field update

Suppressing the parameter argument, HERMES propagates the three marginals.

$$\begin{aligned} \mu_{k,t,u} &:= \prod_{v \in \mathcal{N}_u} (1 - \beta^I f_{k,t-1,v}^I) \\ f_{k,t,u}^S &:= f_{k,t-1,u}^S \mu_{k,t,u} \\ f_{k,t,u}^I &:= (f_{k,t-1,u}^I + f_{k,t-1,u}^S (1 - \mu_{k,t,u})) (1 - \beta^R) \\ f_{k,t,u}^R &:= 1 - f_{k,t,u}^S - f_{k,t,u}^I \end{aligned}$$

The marginals at each right endpoint score the observed node states.

Joint objective

We maximize this objective over all observed snapshots.

$$\hat{\beta} := \arg \max_{\beta \in (0,1)^2} \sum_{k=1}^K \sum_{u \in \mathcal{V}} \log f_{k,t_k,u}^{y_{t_k,u}}(\beta)$$

HISTORY RECONSTRUCTION VIA MCMC

Hitting-time representation

The first infection and recovery times compactly encode each node's monotone $S \leq I \leq R$ trajectory.

$$\begin{aligned} h_u^I(\mathbf{Y}) &:= \min\{T+1, \min\{t \geq 0 : y_{t,u} \in \{I, R\}\}\} \\ h_u^R(\mathbf{Y}) &:= \min\{T+1, \min\{t \geq 0 : y_{t,u} = R\}\} \end{aligned}$$

Posterior barycenter

Using hitting times as coordinates, we reconstruct the squared-distance barycenter of the posterior. Bias–variance decomposition yields an exact closed-form solution.

$$h_u^x(\hat{\mathbf{Y}}) := \text{round}\left(\mathbb{E}_{\mathbf{Y} \sim P_{\hat{\beta}}(\cdot | \mathbf{y}_{\mathcal{T}_{\text{obs}}})}[h_u^x(\mathbf{Y})]\right), \quad x \in \{I, R\}$$

M–H posterior sampling

Direct posterior sampling is intractable. Draw a candidate from the feasibility-aware proposal and apply the M–H correction.

$$\alpha(\mathbf{Y}, \tilde{\mathbf{Y}}) := \min\left\{1, \frac{P_{\hat{\beta}}[\tilde{\mathbf{Y}}] Q_{\theta}(\mathbf{y}_{\mathcal{T}_{\text{obs}}})[\tilde{\mathbf{Y}}]}{P_{\hat{\beta}}[\mathbf{Y}] Q_{\theta}(\mathbf{y}_{\mathcal{T}_{\text{obs}}})[\mathbf{Y}]}\right\}$$

The resulting Markov chain targets the desired posterior distribution; averaging its hitting times and rounding them reconstructs the complete history $\hat{\mathbf{Y}}$.

FEASIBILITY-AWARE PROPOSAL

M–H MCMC requires a proposal that is tractable to sample from while remaining compatible with every observed snapshot.

Segment-wise proposal

Observed snapshots decompose the proposal into interval-wise factors.

$$\begin{aligned} Q_{\theta}(\mathbf{y}_{\mathcal{T}_{\text{obs}}})[\mathbf{Y}] &= \\ &\times \prod_{k=2}^K Q_{\theta}(\mathbf{y}_{\mathcal{T}_{\text{obs}}})[\mathbf{Y}_{[0,t_1]} | \mathbf{y}_{t_1}] \\ &\times \prod_{k=2}^K Q_{\theta}(\mathbf{y}_{\mathcal{T}_{\text{obs}}})[\mathbf{Y}_{(t_{k-1},t_k]} | \mathbf{y}_{t_{k-1}}, \mathbf{y}_{t_k}] \end{aligned}$$

Reverse-time generation

Generate each segment backward; allow only states that pass the full feasibility test.

$$\mathcal{X}_{t,u} := \{x \in \mathcal{X} : y_{t,u} = x \text{ is feasible}\}$$

Dynamic reachability mask

This enforces both right-end feasibility and left-end extendability.

$$\mathcal{V}_t := \{u \in \mathcal{V} : y_{t_{k-1},u} \neq R, y_{t,u} \neq S, y_{t+1,u} \neq S\}$$

$$\mathcal{W}_t := \{u \in \mathcal{V}_t : \exists v \in \mathcal{V}_t(y_{t_{k-1}}), d_{\mathcal{V}}(v,u) \leq t - t_{k-1}\}, \quad k \geq 2$$

$$k = 1 : \mathcal{W}_t := \mathcal{V}_t$$

Support guarantee

The theoretical proposal support exactly matches the feasible posterior support.

$$\text{supp}(Q_{\theta}(\mathbf{y}_{\mathcal{T}_{\text{obs}}})) = \text{supp}(P | \mathbf{y}_{\mathcal{T}_{\text{obs}}})$$

GNN proposal

An edge-gated GNN predicts two node-wise probabilities.

$$q_{t,u}^I, q_{t,u}^S$$

Training uses simulated histories under the estimated parameters:

$$\min_{\theta} \mathbb{E}_{\mathbf{Y}' \sim P_{\hat{\beta}}}[-\log Q_{\theta}(\mathbf{y}'_{\mathcal{T}_{\text{obs}}})[\mathbf{Y}']]$$

Complexity

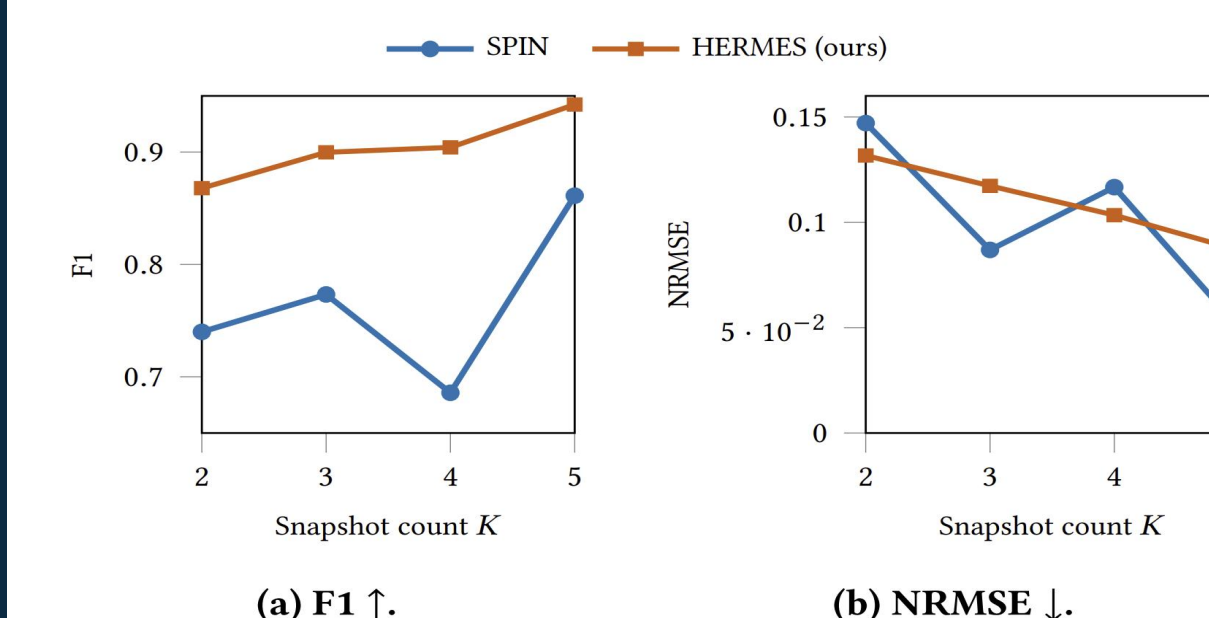
Sampling a complete proposal history costs $O(Tn(n+m))$

MAIN EXPERIMENTS

HERMES: best F1 11 / 12 datasets			HERMES SI avg. rank 1.75		HERMES SIR avg. rank 1.50	
SI / SI-like datasets			SIR / SIR-like datasets			
Dataset	F1 ↑	NRMSE ↓	Dataset	F1 ↑	NRMSE ↓	
BA	GRIN 0.8939 → 0.9012	SPIN 0.1377 → 0.1284	BA	GRIN 0.8633 → 0.8640	GRIN 0.0981 → 0.1509	
ER	SPIN 0.8996 → 0.9011	SPIN 0.1291 → 0.1269	ER	GRIN 0.8631 → 0.8657	GRIN 0.0962 → 0.1307	
Oregon2	SPIN 0.8941 → 0.8964	SPIN 0.1328 → 0.1380	Oregon2	GRIN 0.8706 → 0.8719	GRIN 0.1034 → 0.1874	
Prost	CRI 0.8523 → 0.9056	CRI 0.1984 → 0.1308	Prost	DITTO 0.7929 → 0.8609	DITTO 0.1690 → 0.1737	
BrFarmers	GRIN 0.9043 → 0.8980	GRIN 0.1268 → 0.1491	Covid	SPIN 0.7656 → 0.7819	GRIN 0.1647 → 0.1589	
Pol	GRIN 0.8477 → 0.8487	GRIN 0.1752 → 0.1823	Hebrew	DHREC 0.7718 → 0.8356	CRI 0.1466 → 0.1223	

More observations help.

Increasing snapshots from K=2 to K=5 improves HERMES F1 from **0.8679** to **0.9423** and lowers NRMSE from **0.1317** to **0.0868**.



Learned proposal helps.

On BrFarmers, proposal training improves F1 from **0.8399** to **0.8979** and lowers NRMSE from **0.1888** to **0.1510**.

